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Microbial diversity regulates mercury cycling in paddy soils

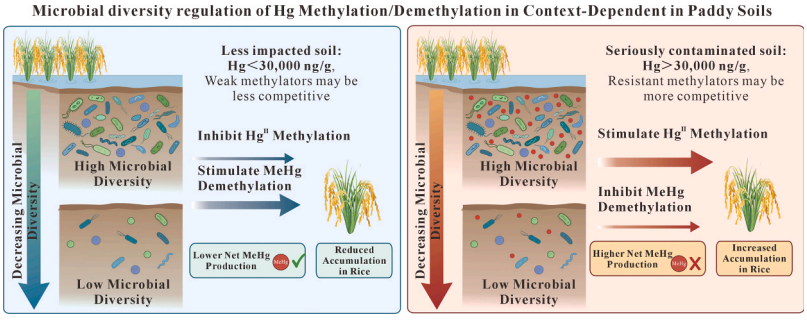
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HIGHLIGHTS

- Combining Hg isotope addition with microbial diversity gradient dilution links microbial diversity to Hg transformation.
- Microbial diversity loss shows opposite effects on methylmercury production.
- A soil Hg threshold (~30,000 ng/g) dictates the direction of microbial effects.
- Findings are validated by an expanded nationwide paddy soil dataset.

GRAPHICAL ABSTRACT



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ABSTRACT

Microbial mercury (Hg) methylation and methylmercury (MeHg) demethylation critically govern MeHg production in paddy soils and its accumulation in rice. However, how the decline in microbial diversity under climate change affects these processes remains unclear. Here, we combined stable isotope tracing with a dilution-to-extinction approach to manipulate microbial diversity across paddy soils with Hg contamination gradient (HX: 165 ng/g, GX: 20,707 ng/g and SK: 659,303 ng/g), investigating its effects on Hg methylation and demethylation. Results showed that diversity loss suppressed methylation (to 0.31–0.82 times the original soil) while enhanced demethylation (to 1.33–7.00 times the original soil) in HX and GX soils, reducing net MeHg production. Conversely, in SK soil, it promoted methylation (to 0.31–0.71 times the original soil) and inhibited demethylation (0.14–0.48 times the original soil), increasing MeHg accumulation. Marginal density curves and linear regression analyses indicate that the regulatory effect of microbial diversity on MeHg production depends strongly on Hg levels, with a significant shift in MeHg concentration across a critical threshold of 30,000 ng/g. Nationwide expanded data further confirmed that diversity loss deceases MeHg production below this threshold but elevates it above. This divergence is attributed to Hg-induced shifts in microbial community structure

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induced. Our findings highlight the crucial role of microbial diversity in regulating net MeHg production in paddy soils, offering important insights for predicting Hg risks under climate change and ensuring food security.

1. Introduction

Mercury (Hg) contamination remains a pressing global-change issue due to its high persistence, long-range atmospheric transport, and potent neurotoxicity [1]. In rice-based agroecosystems, paddy soils represent one of the most important sources of methylmercury (MeHg)-the bioaccumulative and far more toxic organic form of Hg [2–5]. Unlike other crops, rice can accumulate substantial MeHg in edible grains, making it a dominant MeHg exposure pathway for populations in Asia and increasingly in other rice-producing regions [2,6–10]. Paddy soil, as a typical constructed wetland, is the primary source of MeHg in rice grain [3,11]. As global food demand continues to rise and rice cultivation expands, understanding the controls on MeHg production in paddy soils is becoming ever more critical for food security and human health.

Net MeHg production is controlled by the leverage between Hg methylation and MeHg demethylation [12–14]. Microorganisms mediate virtually almost all Hg methylation and much of MeHg demethylation in soils [15,16]. Microbial Hg methylation is primarily governed by the *hgcAB* gene cluster, while MeHg demethylation occurs via reductive (mediated by the *mer* operon) or oxidative pathways [17,18]. Importantly, non-methylating microorganisms also play a crucial role by modulating nutrient competition, redox conditions, and maintaining the overall community stability required for these transformations [19]. These microbial processes are regulated by a complex interplay of microbial functional groups, organic matter transformations, and redox-sensitive sulfur and iron cycling. While environmental factors such as dissolved organic matter, redox dynamics, sulfur speciation, and Hg bioavailability have been extensively studied as drivers of Hg methylation/demethylation [15,20–22], the specific role of microbial diversity has received comparatively little attention. While previous studies have established that Hg contamination can significantly alter microbial diversity [23], the relationship between microbial diversity and Hg cycling is likely interactive. Given the mounting evidence that microbial diversity is a critical regulator of ecosystem multifunctionality and biogeochemical cycling in natural environments [24–26], understanding how microbial diversity regulates Hg cycling is of particular importance.

Soil microbial diversity is highly sensitive to environmental perturbations [27,28]. Global climate change is projected to exert strong and pervasive impacts on soil microbial diversity through warming, altered precipitation regimes, and intensified drought-flood cycles [29–31]. Warming experiments and long-term observational studies consistently show that climate change reduces microbial diversity by filtering out less-adapted taxa and restructuring community composition [32,33]. Decreasing microbial diversity can significantly affect the cycling of biogenic elements. This includes reduced carbon storage efficiency [34], impaired nitrification potential, decreased soil nitrate and methane flux [35,36], and significantly inhibiting specific functional capabilities (e.g. pesticide degradation, etc.) [37]. The decreasing microbial diversity may reduce or alter the key functional taxon (such as SRB, IRB) related to Hg transformations, and simultaneously affect the expression of functional genes (such as *hgcA*, *merB*), thereby interfering with the dynamic balance of Hg methylation and MeHg demethylation. This imbalance will further affect the net MeHg production, the absorption and accumulation of MeHg by rice, and ultimately influence the exposure risk of MeHg intake through rice consumption in the population.

Existing studies provide indirect hints that microbial diversity could modulate Hg methylation, but they focus on individual taxa (e.g., *Geobacter*, *Anaerolinea*, *Dechloromonas denitrificans*) or specific functional genes (e.g., *hgcA*), rather than the broader community context [38–40]. Although the effects of microbial diversity may regulate Hg transformation through changes in the abundance of known methylators, shifts in community assembly processes, or interactions with physicochemical conditions such as sulfur speciation, it remains unclear whether a decrease in microbial diversity itself suppresses or enhances Hg methylation and MeHg demethylation. More importantly, whether these responses vary across contamination gradients is still not known. To address these knowledge gaps, we combined a controlled microcosm experiment manipulating microbial diversity with multi-isotope ($^{202}Hg^{II}$ and $Me^{198}Hg$) tracing to directly quantify Hg methylation and MeHg demethylation processes across a Hg contamination gradient. We further integrated these results with an expanded field dataset to establish a mechanistic link between microbial community structure and Hg methylation/demethylation processes, and to assess the potential impacts of microbial diversity loss on net MeHg production in paddy soils. By linking microbial ecology with Hg biogeochemistry, our study provides mechanistic insights into how climate-driven microbial diversity loss reshape MeHg accumulation in paddy soils, with broad implications for food safety and environmental risk under global change.

2. Materials and methods

2.1. Site description and sampling

Paddy soils were collected from three sites in Guizhou Province, Southwest China, representing a gradient Hg contamination: an abandoned Hg mining area (SK), an artisanal Hg smelting site (GX), and an uncontaminated rural site near Guiyang City (HX). Detailed site characteristics, including THg and MeHg concentrations in the soils are provided in Text S1. Surface soils (1–10 cm below the soil—water interface) were sampled on 50–60 days after rice plant transplantation (August 2023). This continuously flooded stage is known to favor highly active Hg methylation in paddy systems, during which soil MeHg concentrations have been reported to be up to 8 times higher than those during drained periods [13,14]. At each site, soils from four random locations within 500 m^2 were homogenized to obtain one composite sample. Samples were collected into 500 mL sterile polypropylene bottles without headspace, sealed with Parafilm®, double-bagged, and transported at 4°C for incubation.

2.2. Microcosm incubation experiment design

Anaerobic incubation experiments were performed in an oxygen-free (N_2-filled) glovebox (PLAS-LABS, USA). Following established methodology [35], a stock soil suspension was prepared by mixing 50 g of paddy soil with 450 mL of sterilized deionized water (121 ℃ for 30 min, Milli-Q®, Millipore, USA) to create a T1-gradient suspension. Serial dilutions were then prepared from this stock to generate the inocula for subsequent treatments. Five distinct treatments were established: original soil (OS, containing 20 g non-sterilized soil and 30 mL sterilized water); sterilized soil control [20 g of sterilized paddy soil (0.6 h^{-1} gamma irradiation for 48 h) and 30 mL of sterilized deionized water], a dose demonstrated to effectively eliminate soil microorganisms while optimally preserving physicochemical properties (especially organic matter) compared to harsh autoclaving [41,42]; and three inoculation treatments (T1–T3), each containing 20 g sterilized soil amended with 30 mL of serial dilutions (T1, T2, and T3 gradients, respectively) from the stock suspension (Fig. 1a). All treatments were pre-incubated at 15°C for one week to activate anaerobic microorganisms [43].

Each treatment was conducted in triplicate, totaling 45 microcosms (3 sites × 5 treatments × 3 replicates). Enriched Hg isotope tracers, including $^{202}Hg(NO_3)_2$ ($^{202}Hg^{II}$) and $Me^{198}HgNO_3$ ($Me^{198}Hg$) were

spiked simultaneously into the same soil slurries to trace Hg methylation and MeHg demethylation, respectively. The amounts of spiked $^{202}Hg^{II}$ and $Me^{198}Hg$ tracers were ~10% and ~100% (Table S1), respectively, of the soil ambient total Hg (THg) and MeHg concentrations [12,44]. More details on the preparation of the $^{202}Hg^{II}$ and $Me^{198}Hg$ tracers are available in our previous study [45]. Given that simultaneous incubation of soil samples from different sites may lead to contamination of uncontaminated soil samples, we quickly closed the serum bottle after isotope addition and moved it outside the glove box, followed by cleaning the glove box. In addition, the incubation order was carried out from low to high according to the ambient THg concentration (i.e. HX: 165 ng/g-GX: 20,707 ng/g-SK: 659,303 ng/g).

Three bottles were destructively sampled in the glovebox after 24 h of incubation due to the systematic underestimation of rate constants over longer durations [46]. Samples for the analysis of isotopic MeHg concentrations and DOM were collected into 50 mL PP tubes (JET®, BIOFIL, China), stored at −20°C, freeze-dried and ground evenly through a 200-mesh sieve using an agate mortar. To investigate the impact of alterations in microbial community structure on Hg methylation and MeHg demethylation, soil slurry incubation samples for 16S rRNA gene sequencing were collected into cryogenic tubes (2 mL, Nalgene®, Thermo Scientific, USA) and stored at −80°C for further analysis. Sulfide and iron were determined immediately (as both species are redox sensitive) using methylene blue method and ferrozine method, respectively.

2.3. Expanded dataset and statistical modeling

To generalize the relationship between microbial diversity and Hg methylation beyond the incubation experiment, we compiled an expanded dataset comprising measured MeHg concentrations and microbial diversity indices (Shannon Index) from multiple published studies [47–51] and a recent nationwide survey (including 21 newly collected samples). The nationwide dataset comprised 250 soil samples collected from major rice-growing regions across eight provinces in China (Guizhou, Hubei, Hunan, Anhui, Shaanxi, Jiangsu, Liaoning and Chongqing). The samples represented a wide gradient of Hg contamination, with THg concentrations ranging from 6.30 ng/g to 1079, 748 ng/g (Fig. S1 and 5a). Sampling procedures for the nationwide samples followed the methodologies described in Section 2.1.

We performed two complementary analyses: (1) Partial Dependence Plot (PDP) analysis using a Random Forest model to visualize the marginal effect of microbial diversity on Hg methylation, with LOESS smoothing (span=0.75) applied for trend illustration; (2) Third-order polynomial regression to quantify the relationship between microbial diversity and MeHg concentration, with model fit assessed by R^2 values.

2.4. Analytical methods

2.4.1. Methylation and demethylation rate constants

Gas chromatography-inductively coupled plasma—mass spectrometry (GC-ICP—MS, Agilent 7700x, Agilent Technologies, Inc., USA) followed by the ethylation-purge trap method was employed to

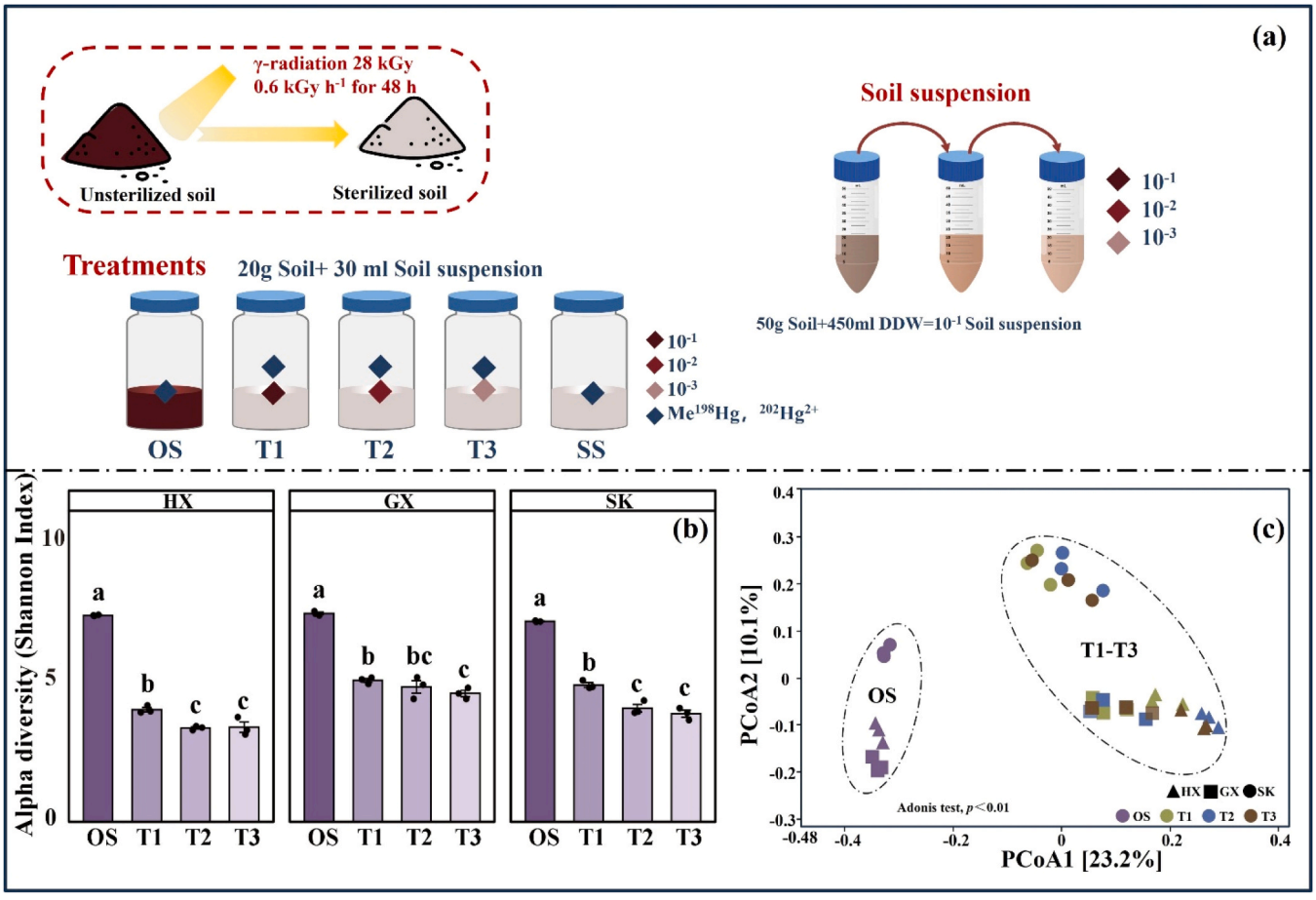


Fig. 1. Experimental design and variation in microbial diversity and community in different treatments. (a) Schematic diagram of the experimental design for microbial diversity manipulation and anaerobic incubation. **(b)** The alpha-diversity based on Shannon index. Different lowercase letters indicate that the differences between the treatments at different microbial diversity are significant (one-way ANOVA, $p < 0.05$). **(c)** The beta-diversity indicated by principal coordinate analysis (PCoA) based on unweighted UniFrac distances.

determine MeHg isotopes [44,52]. Hg methylation was monitored via the formation of $Me^{202}Hg$ from spiked $^{202}Hg^{II}$, while MeHg demethylation was assessed by the decrease in spiked $Me^{198}Hg$ over time [44,52]. The first-order methylation rate constant (K_m) and demethylation rate constant (K_d) were calculated using Eqs. (1) and (2), respectively [44,52]. More details of the Hg isotope analysis can be found in our previous works [45,51].

$$K_m = \frac{[Me^{202}Hg]_{ti} - [Me^{202}Hg]_{t0}}{[^{202}Hg^{II}] \times (t_i - t_0)} \quad (1)$$

$$K_d = \frac{\ln\left([Me^{198}Hg]_{t0}\right) - \ln([Me^{198}Hg]_{ti})}{(t_i - t_0)} \quad (2)$$

Here, $[Me^{202}Hg]_t$ and $[Me^{198}Hg]_t$ are the $Me^{202}Hg$ and $Me^{198}Hg$ concentrations, respectively, which were evaluated initially (t_0) and after 24 h (t_i); $[^{202}Hg^{II}]$ is the $^{202}Hg^{II}$ concentration in the initial spike added to the soils. More calculation details are provided in Text S2.

2.4.2. Physicochemical parameters

The liquid phase of incubation soil slurry was separated by centrifugation (RCF = 2850 g for 10 min at 4°C) under anaerobic conditions (in N_2-filled glovebox) for sulfide (S^{2-}) and iron (Fe^{2+} and Fe^{3+}) analysis, immediately (as both species are redox sensitive). The sulfide (S^{2-}) concentration in the aqueous phase of the soil slurry was determined using the Cline method [53]. The detection limits for sulfide analysis are 0.13 μM [53]. Ferrous (Fe^{2+}) and ferric (Fe^{3+}) cations were measured using the ferrozine method [54]. The pH values of the soil samples were measured using a pH electrode (SX751 pH/ORP/Cond/DO meter, SANXIN, China). The detailed testing methods for THg and MeHg in soil background are provided in Text S3.

2.4.3. Characterization of microbial communities

Total microbial DNA in slurry samples was extracted from 0.5 g of soil using FastDNA® Spin Kits (MP Biomedicals, Irvine, CA) according to the manufacturer's protocol. The concentration and quality of the DNA were determined using the fluorescent method (Qubit™ 4 Fluorometer, Thermo Scientific, USA) [23]. To characterize the soil microbial communities, the V3-V4 region of the microbial 16S rRNA gene was amplified using the universal bacterial primers 515F (5′-GTGYCAGCMGCCGCGGTAA-3′) and R806 (5′-GGACTACNVGG GTWTCTAAT-3′). More details on the 16S rRNA gene sequencing procedure are available in Text S4. Microbial Hg methylation genes (*hgcA*) MeHg demethylation genes (*merB*) were determined by real-time quantitative PCR (qPCR) on the ABI7500 (Applied Biosystems, USA); details are provided in Table S2.

2.5. QA/QC and statistics

Quality control for MeHg isotope analysis of the soil samples was determined using blanks, triplicate samples, and certified reference material (CRM, CC580). The method detection limit (3σ) for MeHg isotope analysis was 0.015 ng/g. The relative standard deviation (RSD %) of triplicates was < 15%. The data are presented as the mean ± standard error (SE). The MeHg recovery from CRM ranged from 80% to 93%, with an average of 86 ± 4% (n = 13). The one-way ANOVA (LSD post hoc test) was employed to determine statistically significant differences ($p < 0.05$,2-tailed) between treatments. All statistical computations and visualizations were performed within the R software environment. This included principal coordinate analysis (PCoA) of unweighted UniFrac distances, the Adonis test, and calculations of microbial alpha diversity, which were conducted using the "vegan" package [55,56]. The relative importance of deterministic versus stochastic processes in microbial community assembly was quantified for each sample by calculating the Nearest Taxon Index (NTI) with the "picante" package using 999 randomizations. To identify the significant drivers of the methylation/demethylation potential parameters (K_m and K_d), a random forest model was constructed using the "RandomForest" package [57]. For the expanded dataset, partial dependence plots (PDPs) derived from the Random Forest models were applied to visualize the marginal effect of microbial diversity on Hg methylation potential, and their relationship was further quantified using third-order polynomial regression. Furthermore, Partial Least Squares Structural Equation Modeling (PLS-SEM) was performed using the 'plspm' package [58,59], where the statistical significance of the path coefficients was evaluated via a bootstrapping procedure with 5000 resamples. The data analysis workflow was supported by the integrated use of the tidyverse, caret, pdp, and ggplot2 packages.

3. Results

3.1. Microbial diversity in the microcosms

The soil microbial community was modulated using a gradient dilution method, resulting in distinct microbial diversity and composition across treatments (Fig. 1). Alpha diversity, as measured by the Shannon index, was highest in the original soil at the HX (7.32 ± 0.02), the GX (7.39 ± 0.08), and the SK (7.10 ± 0.02), but significantly decreased with increasing dilution in treatments T1 (HX: 3.98 ± 0.13; GX: 5.02 ± 0.11; SK: 4.85 ± 0.15), T2 (HX: 3.33 ± 0.07; GX: 4.78 ± 0.39; SK: 4.03 ± 0.24), and T3 (HX: 3.35 ± 0.32; GX: 4.55 ± 0.20; SK: 3.83 ± 0.21) ($p < 0.05$; Fig. 1b). Principal coordinates analysis (PCoA) based on unweighted UniFrac distances revealed significant differences in bacterial community composition across the diversity gradient (Adonis, $p < 0.01$; Fig. 1c). The abundance of the *hgcA* (log10-transformed copies) varied distinctly among sites (Fig. S2). In HX, *hgcA* gene copies in T1, T2, and T3 were significantly lower than in original soil by 0.03–0.24 times ($p < 0.05$), with the greatest reduction observed in T3 (Fig. S2). Similarly, in GX, *hgcA* gene copies decreased across treatments, though the differences were not statistically significant ($p > 0.05$). In contrast, SK exhibited an opposite trend, with *hgcA* gene copies in T1, T2, and T3 significantly higher than that in original soil by 0.02–0.16 times ($p < 0.05$) (Fig. S2). Overall, these results demonstrate that dilution-induced reductions in microbial diversity were accompanied by clear shifts in *hgcA* gene abundance, though the direction and magnitude of change were site-specific. Copies of the *merB* were below the detection limit in all treatments throughout the sampling sites [real-time threshold cycle (C_T) > 31].

3.2. Hg methylation and MeHg demethylation in the microcosms

The methylation rate constant (K_m-$^{202}Hg^{II}$) and methylmercury ($Me^{202}Hg$) concentrations produced from spiked $^{202}Hg^{II}$ in sterilized soil were significantly lower than those in unsterilized treatments (Original soil, T1, T2, T3) across all sites ($p < 0.05$), indicating that microbial processes predominantly drive Hg methylation in paddy soils, with negligible abiotic contributions (Fig. 2a and Fig. S3). With decreasing microbial diversity, K_m-$^{202}Hg^{II}$ exhibited contrasting trends among sites (Fig. 2a). In the HX and GX, K_m-$^{202}Hg^{II}$ decreased significantly in T1, T2, and T3 compared to original soil ($p < 0.05$), with reductions ranging from 0.31 to 0.82 times when compared with original soil. Specifically, the largest decrease was observed in T3 at GX (0.82 times original soil). It is worth noting that at the GX site, K_m-$^{202}Hg^{II}$ in the T2 exhibited an anomalous value higher than those in T1 and T3; similarly, at the HX site, the K_m-$^{202}Hg^{II}$ in the T3 slightly deviated from the declining trend observed between T1 and T2 (Fig. 2a). In contrast, in the SK, K_m-$^{202}Hg^{II}$ increased significantly with decreasing microbial diversity ($p < 0.05$), with T1, T2, and T3 showing increases of 0.31–0.71 times relative to original soil, peaking in T3 (0.71 times original soil). $Me^{202}Hg$ concentrations followed similar trends (Fig. S3). This site-specific response difference indicates that the influence of microbial diversity on Hg

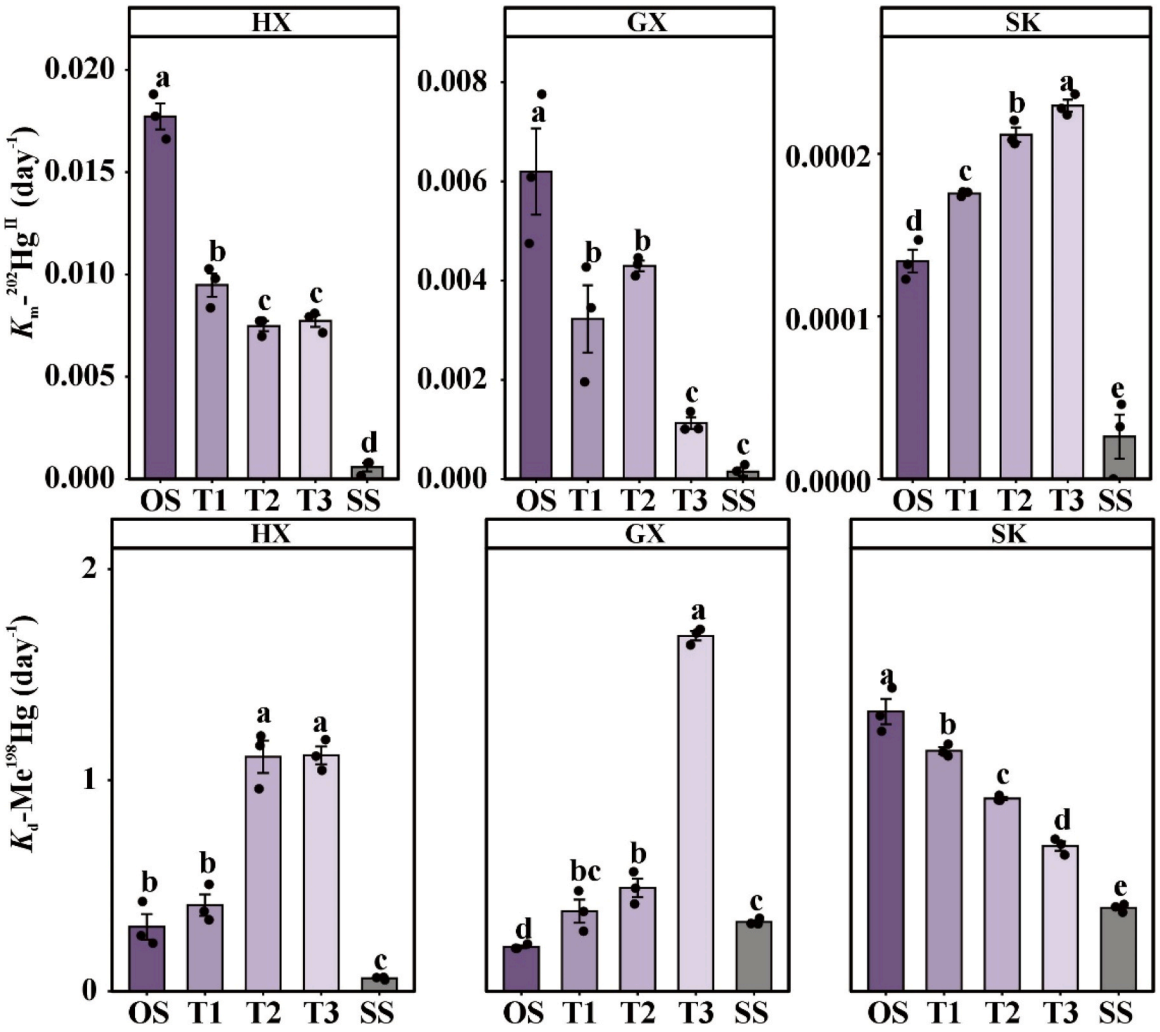


Fig. 2. Hg^{II} methylation/demethylation rate constants for all sites at different microbial diversity gradients. (a) Plot for the methylation rates (K_m-$^{202}Hg^{II}$) for different treatments at all sites. **(b)** Plot for the MeHg degradation rates (K_d-$Me^{198}Hg$) for different treatments at all sites. Different lowercase letters indicate that the differences between the treatments at different microbial diversity are significant (one-way ANOVA, $p < 0.05$).

methylation is not consistent. It strongly depends on the Hg pollution stress that the soil has experienced and the specific microbial community background formed thereby.

The demethylation rate constant (K_d-$Me^{198}Hg$) and $Me^{198}Hg$ concentrations in sterilized soil were significantly lower than in unsterilized treatments (original soil, T1, T2, T3) at HX ($p < 0.05$; Fig. 2b and S4), confirming that microbial processes primarily drive MeHg demethylation in paddy soils. However, K_d-$Me^{198}Hg$ in sterilized soil at GX and SK was higher than that at HX ($p < 0.05$; Fig. 2b), suggesting site-specific abiotic demethylation contributions, which are further explored in Section 4.1. With decreasing microbial diversity, K_d-$Me^{198}Hg$ exhibited divergent trends across sites ($p < 0.05$; Fig. 2b). In the HX and GX, K_d-$Me^{198}Hg$ increased significantly in T1, T2, and T3 compared to original soil, with fold increases ranging from 0.33 to 2.66 at the HX and 1.33 to 7.00 at the GX, peaking in T3 at both sites. In contrast, in the SK, K_d-$Me^{198}Hg$ decreased significantly with decreasing microbial diversity, with T1, T2, and T3 showing reductions of 0.14–0.48 times relative to original soil. This trade-off in microbial ecological function indicates that decreasing microbial diversity stimulates MeHg demethylation in less contaminated environments (e.g. HX and GX), yet suppresses it under high pollution (e.g. SK), highlighting a fundamental reorganization of microbial community functionality.

The net Hg methylation potential (K_m-$^{202}Hg^{II}$/K_d-$Me^{198}Hg$), representing the balance between MeHg production and degradation [14,60], mirrored these trends (Fig. S5). At the HX and the GX, net Hg methylation potential decreased significantly with decreasing microbial diversity ($p < 0.05$), consistent with reduced K_m-$^{202}Hg^{II}$ and increased K_d-$Me^{198}Hg$. Notably, within this declining trend, the net Hg methylation potential for T2 and T3 were very close at the HX; similarly, the net Hg methylation potential for T1 and T2 were close at the GX. Conversely, at the SK, net Hg methylation potential increased significantly ($p < 0.05$), driven by increased K_m-$^{202}Hg^{II}$ and decreased K_d-$Me^{198}Hg$. Background MeHg concentrations (based on $Me^{200}Hg$) followed similar trends, decreasing at HX and GX but increasing at SK with declining microbial diversity ($p < 0.05$; Fig. S6).

3.3. *Community reassembly and functional gene shifts underlie diversity effects on Hg transformation*

Soil physicochemical parameters (DOC, pH, Fe^{2+}, Fe^{3+}, SO_4^{2-} and S^{2-}) exhibited significant variation across the HX, the GX, and the SK sites ($p < 0.001$; Fig. S7). Across microbial diversity treatments (T1, T2, T3), physicochemical parameters remained stable within each site ($p > 0.05$; Fig. 1), suggesting that physicochemical parameters were unaffected by microbial diversity gradients. Analysis of the Nearest Taxon Index (NTI) revealed that dilution treatments induced a significant shift in phylogenetic community structure toward deterministic processes (Fig. S8). However, despite such pronounced community reassembly, no single

microbial taxon showed consistent correlations with net Hg methylation potential across sites (Fig. S9). This suggests that shifts in Hg methylation potential were driven by changes in overall community.

Random Forest modeling confirmed that microbial parameters, including alpha diversity, beta diversity, and *hgcA* gene abundance, served as primary predictors of Hg methylation/demethylation, highlighting the central role of microbial community composition and functional gene (Fig. 3a). These relationships were further supported by a correlation heatmap, which revealed significant associations between microbial diversity, *hgcA* gene copies, and Hg methylation and demethylation rates (K_m and K_d) across all sites (Fig. 3b). Path analysis using PLS-SEM illustrated contrasting mechanistic pathways under different contamination levels (Fig. 4). In the HX and the GX, decreasing microbial diversity reinforced deterministic community restructuring and decreased *hgcA* gene abundance, collectively leading to suppressed net Hg methylation potential (Fig. 4a). In the SK, however, decreasing microbial diversity led to deterministic community restructuring and an increase in net Hg methylation potential (Fig. 4b).

3.4. Generalizability of microbial diversity–MeHg accumulation relationships revealed by an expanded soil dataset

Complementary to the controlled incubation experiments, analysis of the expanded field dataset consistently revealed a significant relationship between microbial diversity and MeHg concentration, thereby confirming the generalizability of the patterns observed in our microcosm experiments (Fig. 5 and Fig. S10). Marginal density curve and linear regression analyses indicated that using 30,000 ng/g of soil total Hg as a threshold, the distribution patterns of MeHg concentration differed significantly on either side of this value (Fig. 5a). This threshold is corroborated by the differential response patterns observed in microcosm experiments between the HX (THg = 165 ng/g) and GX (THg = 20,707 ng/g) sites versus the SK site (THg = 659,303 ng/g). Moreover, this threshold aligns with existing studies on microbial Hg stress response, which document that long-term extreme Hg exposure leads to severe alterations in microbial diversity, community structure, and composition [61–63]. All subsequent analyses of Hg transformation processes were conducted within this grouping framework. In less impacted soil (Hg < 30,000 ng/g), partial dependence plot analysis showed that Hg methylation potential decreased significantly with decreasing microbial diversity (Figs. 5b and 5d). In contrast, seriously contaminated soil (Hg > 30,000 ng/g) exhibited the opposite pattern, with Hg methylation potential increasing significantly with decreasing microbial diversity (Figs. 5c and 5e). Third-order polynomial regression further confirmed these distinct patterns, with less impacted soil following the equation $y = 1.48 + 0.305x - 0.0535x^2 + 0.01x^3$ ($R^2 = 0.22$) and seriously contaminated soil described by $y = -42.2 + 37.3x - 8.75x^2 + 0.621x^3$ ($R^2 = 0.51$) (Figs. 5d and 5e). Together, the congruence between microcosm and field observations demonstrates that microbial diversity regulates Hg methylation potential in a manner dependent on the level of Hg contamination, underscoring the broad applicability and robustness of this ecological mechanism.

4. Discussion

4.1. Microbial diversity regulates Hg methylation and MeHg demethylation

Our integrated analysis provides compelling evidence that microbial diversity serves as an important controller of Hg cycling in paddy soils. The significantly lower Hg methylation rate constant (K_m-$^{202}Hg^{II}$) and MeHg demethylation rate constant (K_d-$Me^{198}Hg$) in sterilized soil compared to unsterilized treatments (Original soil, T1, T2, T3) at HX ($p < 0.05$) underscore that microbial processes dominate Hg methylation/demethylation processes, with abiotic contributions being minimal at the HX (Fig. 2). Although *merB* gene copies were below the detection limit across all treatments, significant microbial demethylation was still observed. This observation aligns well with previous knowledge that *mer*-independent demethylation, mediated by anaerobic consortia such as methanogens and sulfate-reducing bacteria without

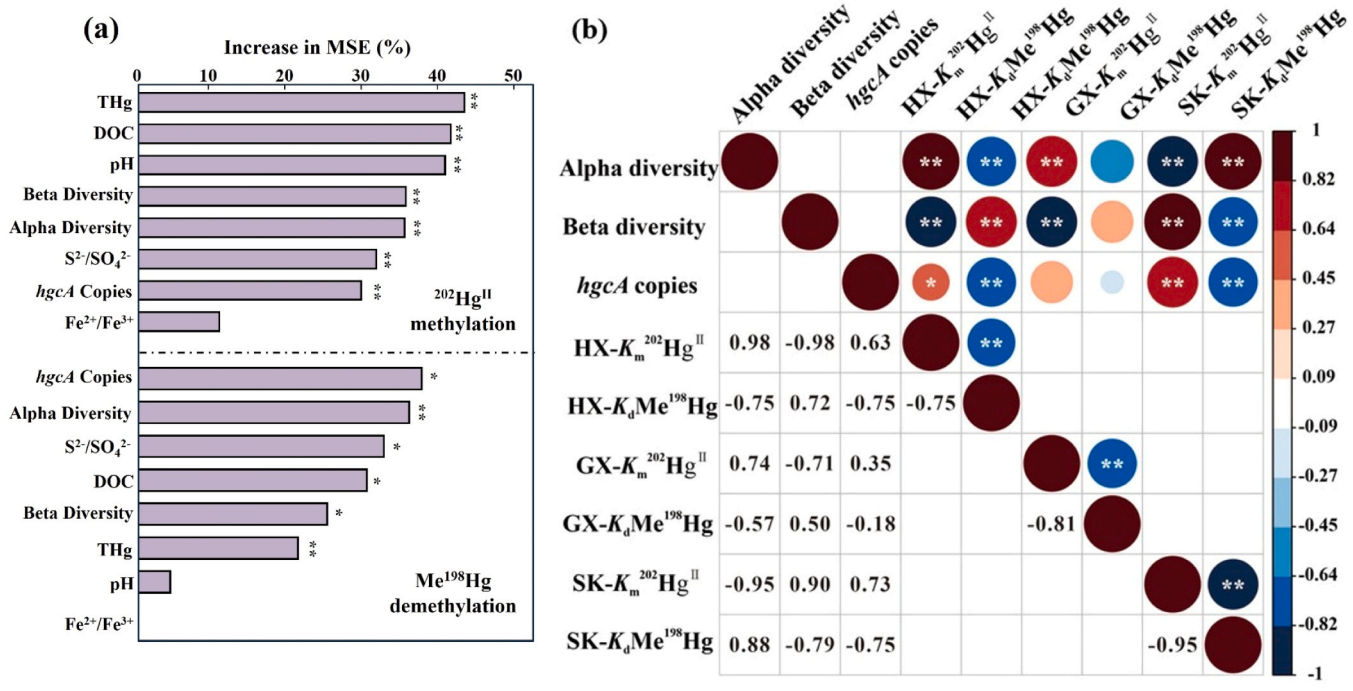


Fig. 3. The contribution of microbial parameters and physicochemical parameters to the Hg^{II} methylation/demethylation with decreasing microbial diversity. (a) The Random Forest model indicating the importance of the predictors for $^{202}Hg^{II}$ methylation and $Me^{198}Hg$ demethylation. **(b)** Heatmap of the correlations among microbial diversity, *hgcA* copies and Hg methylation/demethylation rates. The calculation method is Pearson/Spearman correlation analysis (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

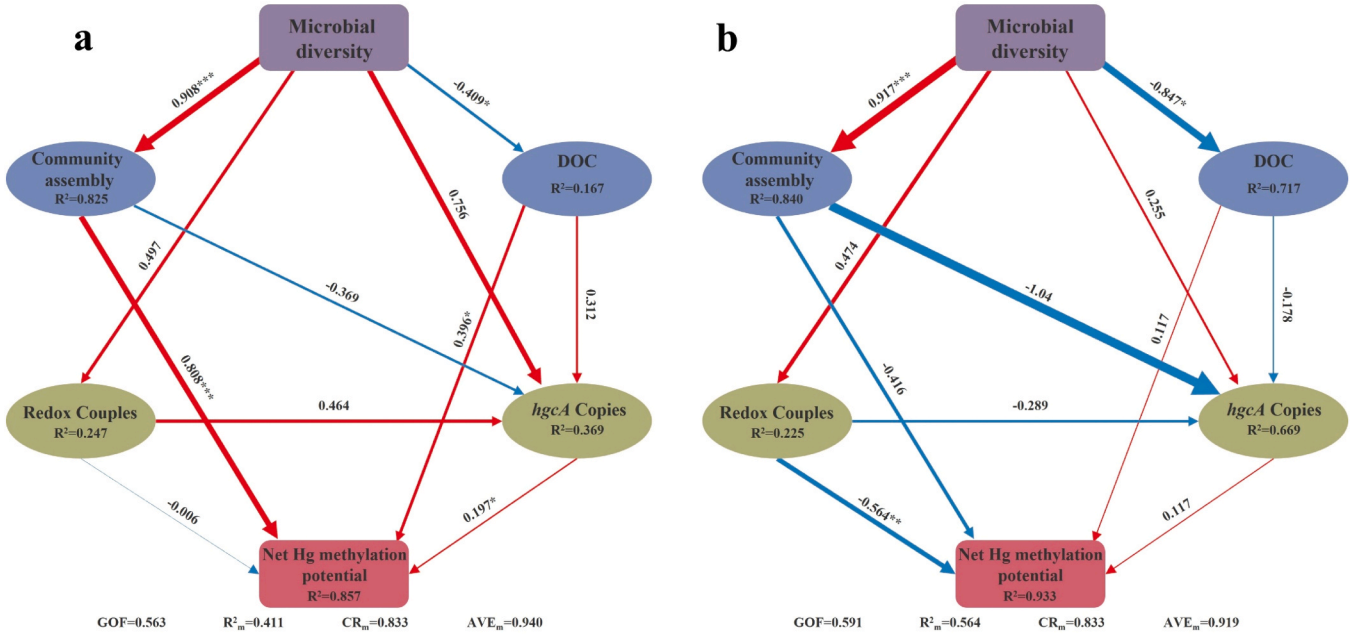


Fig. 4. The path analysis using PLS-SEM illustrated contrasting mechanistic pathways under different contamination levels. (a) less impacted soil. (b) seriously contaminated soil. Redox Couples: Fe^{2+}/Fe^{3+}; S^{2-}/SO_4^{2-}.

the need for the *merB* gene, deeply participates in MeHg degradation in anoxic environments like paddy soils [17]. However, elevated K_d-$Me^{198}Hg$ in sterilized soil at the GX and the SK suggests site-specific abiotic demethylation (Fig. 2b and S4), likely driven by hydroxyl radicals (·OH) generated via possible Fe^{2+}-mediated Fenton reactions following gamma irradiation sterilization, supported by higher Fe^{2+}/Fe^{3+} concentrations at these sites [50,64]. This abiotic contribution, while notable, is secondary to microbial activity, as unsterilized treatments exhibited significantly higher rates across all sites.

Crucially, we discovered that the functional consequences of microbial diversity loss are context-dependent across Hg contamination gradients. In the HX and the GX, decreasing alpha diversity correlated with decreased K_m-$^{202}Hg^{II}$ (0.31–0.82 times Original soil) and increased K_d-$Me^{198}Hg$ (0.33–7.00 times Original soil), reducing net Hg methylation potential (K_m-$^{202}Hg^{II}$/ K_d-$Me^{198}Hg$; Fig. S5). In contrast, in the SK, diversity loss increased K_m-$^{202}Hg^{II}$ (0.31–0.71 times Original soil) and decreased K_d-$Me^{198}Hg$ (0.14–0.48 times Original soil), enhancing net Hg methylation potential (Fig. S5). This trade-off represents a fundamental reorganization of microbial functionality along contamination gradients, highlighting that ecosystem responses to decreasing diversity cannot be generalized but are critically mediated by environmental context, particularly Hg contamination pressure. These findings strongly align with recent paradigms emphasizing that MeHg production in paddy soils is regulated by overall microbial diversity and complex bacterial assemblages rather than isolated Hg methylators [19,65]. Specifically, the loss of diversity can deeply disrupt crucial microbial associations, such as syntrophic interactions between Hg-methylating and putative non-Hg-methylating communities, which collectively dictate the net MeHg formation [19]. When these intricate ecological networks are destabilized by diversity reduction, the resulting functional outcomes, whether enhanced or suppressed methylation, depend fundamentally on the baseline community structure shaped by historical Hg stress [65].

4.2. Diversity effects reflect shifts in microbial assembly and functional bottlenecks

The contrasting Hg cycling patterns emerge from distinct microbial assembly mechanisms triggered by diversity loss under different contamination regimes. Phylogenetic analysis (NTI) confirmed that dilution treatments consistently shifted community assembly towards a more deterministic processes across all sites (Fig. S8). However, the functional outcomes of this reassembly diverged markedly.

In the HX and the GX, higher functional redundancy initially buffered ecosystem processes [66–68], but progressive diversity loss gradually depleted Hg-methylating taxa, as evidenced by declining *hgcA* gene abundance (Fig. S2). This depletion of methylating capacity was further amplified by elevated S^{2-} concentrations (Fig. S7), which likely reduced Hg bioavailability through HgS formation [69,70] and may have promoted microbial demethylation over methylation, as sulfate-reducing bacteria (SRB) can prioritize demethylation under high S^{2-} conditions [71,72]. MeHg demethylation rates increased significantly with decreasing diversity (Fig. 2b), though the absence of identified demethylating taxa limits mechanistic clarity for increased K_d-$Me^{198}Hg$ in these soils.

The THg concentrations at the HX and the GX are 165 and 20,707 ng/g, respectively, which are typical of paddy soils from background to contaminated levels (Hg < 30,000 ng/g). Hierarchical clustering revealed high microbial similarity between the HX and the GX (Fig. S13), with communities dominated by taxa such as *Chloroflexi* and *Calditrichota* (Fig. S14), which are commonly found in less impacted soil [23,38,73,74]. These sites represent a broad range of paddy soils (Hg < 30,000 ng/g), indicating that loss of diversity generally reduces MeHg production in such environments.

In stark contrast, historical Hg stress had already selected for a functionally narrowed community in the SK. The distinct microbial community structure at the SK (Fig. S11 and S13), likely shaped by prolonged Hg exposure, appears to enhance the resilience of Hg-methylating taxa while making MeHg-demethylating populations more vulnerable to decreasing diversity [61,62]. At the SK, decreasing diversity enriched Hg-methylating taxa, increasing *hgcA* copies and K_m-$^{202}Hg^{II}$ while suppressing K_d-$Me^{198}Hg$. No significant correlations were observed between the relative abundance of single reported Hg-methylating or demethylating taxa and the net Hg methylation potential (Fig. S8) [75,76]. In summary, while *hgcA* gene abundance is the direct biological determinant of Hg methylation potential, its variation during environmental disturbance appears to be regulated by the broader microbial community context. Our results suggest a hierarchical

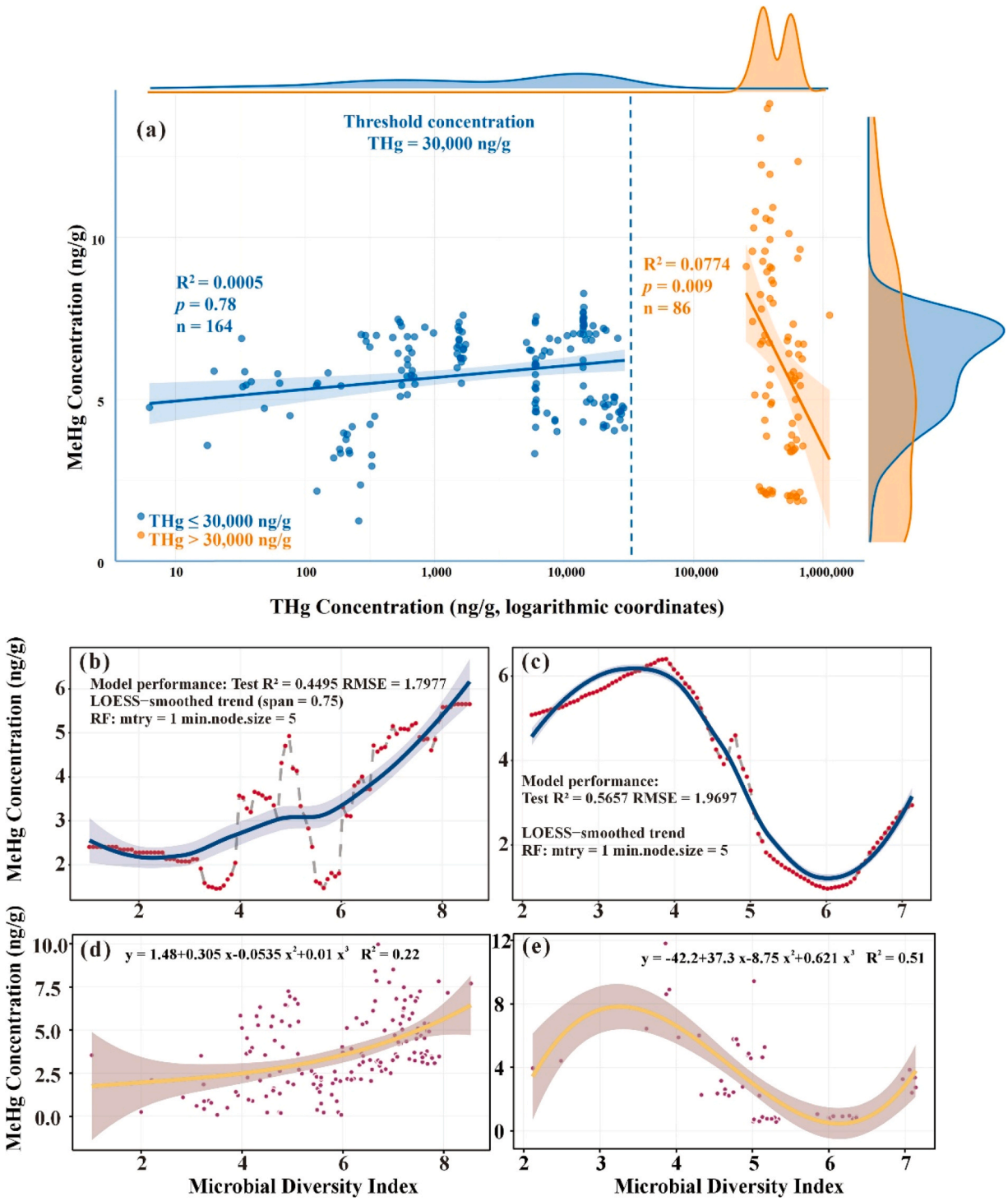


Fig. 5. Contrasting patterns of mercury methylation in relation to a critical soil THg threshold and microbial diversity. (a) Marginal density curves and linear regression fitting revealed distinct distribution patterns of MeHg concentration across different THg concentration ranges. Partial dependence plot (PDP) from Random Forest model showing the marginal effect of microbial diversity (Shannon Index) on MeHg concentration in (b) less impacted soil and (c) seriously contaminated soil. Third-order polynomial regression fits between microbial diversity and MeHg concentration for (d) less impacted soil and (e) seriously contaminated soil.

influence: the initial Hg contamination level shapes the baseline community resistance, which then determines whether the loss of microbial diversity suppresses or stimulates the relative abundance of *hgcA*--harboring taxa.

4.3. *Nonlinear patterns across microcosms and field soils*

The relationships between microbial diversity and Hg methylation observed in our incubation experiments are robustly supported by independent analyses of large-scale field data. Analysis of an expanded dataset was conducted using 30,000 ng/g Hg as a critical threshold, based on marginal density curves and linear regression, which indicated

distinctly different distribution patterns of MeHg concentration on either side of this value (Fig. 5a). This threshold is supported by the differential response patterns observed between the HX/GX and SK sites in microcosm experiments (Fig. 2), and aligns with existing studies showing that high-level Hg exposure severely alters microbial community structure [61–63]. National samples were accordingly grouped using this threshold, and further analyzed with Partial Dependence Plots (PDP) and polynomial regression (Fig. 5). The results consistently revealed that the relationship between microbial diversity and MeHg concentration varies systematically with contamination levels. This demonstrates that microbial diversity differentially regulates MeHg production under varying Hg pollution loads, confirming that Hg contamination level is a key environmental factor influencing microbially mediated Hg methylation. In less impacted soil (Hg $<$ 30,000 ng/g), Hg methylation potential decreased with increasing diversity, while the opposite trend was observed in seriously contaminated soil (Hg $>$ 30,000 ng/g) (Fig. 5).

Our results demonstrate that microbial diversity exerts threshold effects on Hg methylation and demethylation processes. This finding places Hg cycling alongside other processes known to be influenced by diversity, such as methane oxidation and nitrogen cycling [77,78]. The consistency between our controlled experiments and field observations strengthens the conclusion that microbial diversity is a key driver of Hg transformations in paddy soils. Our expanded dataset, which covers a range of environmental conditions, enhances the generalizability of these predictions for natural settings. This work expands upon the important research on specific Hg-methylating microorganisms by examining the role of the broader microbial community [39,40]. Our findings reveal that overall community structure and diversity are critical parameters that co-regulate both Hg methylation and MeHg demethylation. The finding that Hg methylation and MeHg demethylation respond differently to decreasing diversity across sites indicates that the net MeHg outcome is not solely determined by key microbes, but by the functional properties of the entire microbial community.

4.4. Implications for climate change, food safety, and management: limitations and future directions

Climate change is projected to fundamentally reshape soil microbial communities by increasing temperature fluctuations, intensifying drought–flood cycles, and altering nutrient and redox dynamics [30]. These environmental stresses often lead to declines in microbial diversity, with less-adapted taxa disappearing and only a subset of tolerant species persisting [32,79]. Our results demonstrate that such diversity loss can directly alter the balance between Hg methylation and MeHg demethylation, thereby regulating net MeHg production in rice paddies. Specifically, in less impacted soil (Hg $<$ 30,000 ng/g), decreasing diversity suppressed Hg methylation and stimulated MeHg demethylation, leading to reduced MeHg accumulation. In contrast, in seriously contaminated soil, the same process could exacerbate net Hg methylation potential, increasing food safety risks. These contrasting patterns suggest that under climate-driven diversity decline, less impacted soil may experience reduced MeHg risks, whereas seriously contaminated soil, such as those near legacy mining sites, may face exacerbated MeHg accumulation.

Moreover, climate change can indirectly interact with microbial processes by altering soil chemistry [80,81]. Warming accelerates organic matter decomposition, potentially lowering DOC and enhancing Hg bioavailability [82,83]. Fluctuating redox conditions under alternating drought and flooding regimes can shift Fe–S cycling, thereby influencing Hg speciation and the activity of sulfate-reducing and iron-reducing bacteria [84,85]. Together, these mechanisms indicate that climate change will likely magnify site-specific differences in Hg cycling, with high-risk sites becoming even more vulnerable. By linking microbial diversity loss to climate change, our findings highlight a critical knowledge gap: predictive models of MeHg production under global change scenarios must integrate both microbial ecological dynamics and Hg pollution levels. Such integration is essential for anticipating the trajectory of MeHg risks in rice-dominated agricultural systems under future climate conditions.

The implications of this study are twofold. First, protecting microbial diversity, especially in Hg-contaminated areas, should be a priority for reducing MeHg risks in agricultural systems. Practices that maintain or enhance microbial diversity, such as reducing intensive tillage and optimizing fertilizer use, could help mitigate the harmful effects of Hg contamination. Second, climate change adaptation strategies should account for the potential shifts in microbial community dynamics. Targeted interventions, such as sulfur management or redox cycling control, could help regulate Hg transformations in soils vulnerable to Hg contamination, thereby minimizing the risk of MeHg bioaccumulation in rice grains [45,86].

Several key limitations point to future research directions. The precise mechanisms, particularly regarding MeHg demethylation and the role of specific microbial taxa, remain unclear. Future work should employ transcriptomics to directly measure microbial activity, such as *merB* gene expression. Furthermore, controlled experiments are needed to clarify how changes in microbial diversity regulate other factors, including differences in Hg bioavailability, variations in chemical speciation between inorganic Hg and MeHg, and their synergistic effects with Fe–S cycling. Finally, to accurately predict risks under climate change, models must couple these microbial and biogeochemical processes with hydrological and soil organic matter dynamics, and validation should be conducted across seasonal and regional scales.

5. Conclusion

Our study reveals that microbial diversity is a critical, context-dependent regulator of Hg biogeochemistry in paddy soils. Diversity loss produces opposite effects on methylation and demethylation depending on Hg contamination level, driven by shifts in community assembly and functional gene distributions. These findings provide a mechanistic basis for understanding and predicting MeHg dynamics under global environmental change, with important implications for food safety and agricultural management in Hg-affected regions.

Environmental Implication

our study reveals that climate-driven loss of soil microbial diversity can differentially regulate methylmercury (MeHg) production in paddy soils, either mitigating or exacerbating risk depending on historical pollution levels. This contamination-dependent threshold effect demonstrates that microbial diversity is a critical, yet overlooked, factor in mercury biogeochemistry. The findings underscore the necessity of integrating microbial community dynamics into environmental risk assessments and management strategies to accurately predict and mitigate food safety threats under global change.

CRediT authorship contribution statement

Qiang Pu: Writing – review & editing, Methodology, Formal analysis, Data curation, Conceptualization. **Zhengdong Hao:** Conceptualization. **Qianshuo Zhang:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Jiang Liu:** Writing – review & editing. **Kun Zhang:** Writing – review & editing. **Boning Bing:** Writing – review & editing. **Xinbin Feng:** Writing – review & editing, Resources. **Lei Zhao:** Writing – review & editing. **Bo Meng:** Writing – review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2026.142544.

Data availability

Data will be made available on request.

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